

Scientific Publication Indexing and Definition Extraction Resource

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Abstract—Defining relevant concepts from scientific texts is always challenging because of the complex and large amount of information, especially in fast-growing disciplines. We propose SPIDER (Scientific Publication Indexing and Definition Extraction Resource), a tool that can automatically index and define scientific papers in portable document format (PDF) and text format. The primary task of SPIDER is to address the issue of understanding massive scientific documents and further improves the accessibility of scientific information. By applying the state-of-the-art natural language processing (NLP) techniques and self-contained large-language models (LLM) such as Ollama, phi3, and mistral; SPIDER increases the readability of the scientific contents. The model uses the conversion of large PDF document to text, and then it applies NLP methods such as tokenization, stemming, and part-of-speech tagging to create an accurate glossary that is further fine-tuned with the help of self-hosted LLMs for detailed description. The SPIDER method uses NLP modules such as PyPDF2 for PDF handling, NLTK for NLP operations, and transformers for state-of-the-art language modeling to efficiently comprehend scientific articles. The paper describes the SPIDER approach to data cleaning, tokenizer, stemmer, PoS tagger, and the use of self-hosted LLMs. Potentially this model could benefit the enhancement of various fields such as speeding up a literature review, enhancement of educational materials, creating comprehensive glossaries, and supporting the development of knowledge-based systems in scientific and technological disciplines. The result shows SPIDER's effectiveness in generating accurate glossaries achieving optimal performance with a threshold range from 0.1 to 0.4. Thus, SPIDER can be considered a useful tool for researchers, professors, and others who aim to handle a large amount of scientific information.

Index Terms—Automatic-indexing, Scientific paper extraction, Transformers, Natural Language Toolkit, Ollama

I. INTRODUCTION

In the realm of scientific research, the accessibility and comprehension of complex texts are paramount for fostering innovation and advancing knowledge. Researchers frequently encounter the challenge of deciphering specialized terminology and dense technical language within scientific publications. The need for efficient tools to extract and define key

terms has never been more critical, particularly as the volume of scientific literature continues to expand exponentially.

Term extraction is a complex process involving the accurate identification of relevant terms from extensive text corpora, and the difficulty intensifies when dealing with unique terms that do not follow general linguistic patterns. This complexity necessitates advanced methods to discern the precise meanings. The study [1] showcases the challenges of this task and demonstrates how optimizing various automated methods can enhance the precision of extracting domain-specific terms. The setup of earlier automated techniques requires a lot of work, either creating parsing rules, coding excessively, fine-tuning, etc. [2], which could be a monotonous task.

The SPIDER (Scientific Publication Indexing and Definition Extraction Resource) project addresses these challenges by providing an automated solution for the extraction and indexing of definitions from scientific text. SPIDER is designed to streamline the process of generating glossaries and detailed definitions from PDF or text files, thereby enhancing the accessibility of scientific knowledge.

SPIDER employs a robust two-step process. The first step is to convert PDF files to text format and then process the text using various NLP techniques such as tokenization, stemming, and PoS to create an accurate glossary. The second step is to refine this glossary by generating detailed definitions with the aid of self-hosted LLMs (large language models).

SPIDER leverages several powerful modules, including PyPDF2 for PDF processing, NLTK for natural language processing, and transformers for advanced language modeling. By automating the extraction and definition processes, SPIDER significantly reduces the time and effort required to interpret scientific literature, making it an invaluable resource for researchers, educators, and students.

Overall, this paper details the comprehensive methodology for scientific publication indexing and definition extraction. Through SPIDER, we aim to bridge the gap between complex

scientific texts and their understanding, thereby contributing to the democratization of scientific knowledge.

II. LITERATURE REVIEW

In the domain of scientific publication indexing and definition extraction, several notable works have paved the way for advances in automated term extraction and definition generation.

Maciej P. Polak and Dane Morgan [2] involves identifying index pages using keywords and HTML structures, and extracting citation data with lexical, syntactic, and heuristic analyses. Implemented in PubWatcher, this method effectively automates the retrieval of scholarly publications with high accuracy.

Fong, Vu H. L., and Hui S. C. [3] explored effective techniques for automatically extracting scholarly publications from various online sources, including academic websites, digital libraries, and research databases, achieving a high success rate in identifying research papers across these diverse platforms.

Iana Atanassova Marc Bertin and Jean-Pierre Descles [4] presents a method for automatically extracting definitions from scientific texts. The approach focuses on identifying and extracting definitions provided by authors in their papers, particularly those associated with indexed references. The methodology combines linguistic analysis to identify potential definition sentences with a reference-based extraction technique. It leverages indexed citations to accurately pinpoint and extract author-provided definitions from scientific texts, improving the precision of definition extraction in academic literature.

Weijun Fu and Lei Li [5] developed a technique employing Conditional Random Fields (CRF) for automated term extraction. This technique captures the sequential structure of text and utilizes contextual cues to enhance the precision of term detection. It has been notably successful in dealing with domain-specific terminology by taking into account the context in which these terms are situated, thus enhancing the accuracy of the extraction process.

International association members [6] employs a hybrid approach that combines the outputs of three existing information extraction tools: GROBID, ParsCit, and Mendeley. These tools are applied independently to extract header information from research papers, and then their results are integrated using a custom algorithm to produce a more comprehensive and accurate set of extracted metadata elements.

Robert E Mercer and Hospice Hounbo [7] employs a two-step approach: first using linguistic features to extract fine-grained method sentences from a large biomedical corpus, then applying rule-based and machine learning (Conditional Random Field) techniques to extract method terminologies from those sentences.

Liakata, M., Saha, S., Dobnik, S., Batchelor, C. and Rebholz-Schuhmann, D [8] introduce core annotation framework CoreSC and alternative framework AZ-II (Conzat)2 for the primary and secondary annotation of scientific papers,

respectively. CoreSC finds the most general scientific notions, while AZ-II is responsible for the rhetorical structure of scientific arguments. These guidelines have been developed mostly for chemistry papers enabling better insight into the sequence and argumentative of scientific investigations.

Turdakov D., Fedorenko D. and Astrakhantsev N. [9] highlighted the challenges in automatic recognition of domain-specific terms, emphasizing the task due to the unique nature of scientific terminology.

Dominika Tkaczyk and Pawel Szostek [10] introduce a CERMINE system that operates utilizing a modular framework comprising three primary stages: metadata extraction, bibliography extraction, and full-text extraction. This system integrates a spectrum of machine learning methodologies, encompassing both supervised and unsupervised algorithms, to facilitate the processing and extraction of structured data from scientific literature.

Arthur Brack, Jennifer D'Souza, Anett Hoppe, Sören Auer and Ralph Ewerth [11] propose a generic mining task for extracting scientific concepts by introducing a corpus that are annotated with concepts like PROCESS, METHOD, and DATA. The "traditional" scientific literature mining technique will be composed of three sequence-prediction tasks: post-tagging, relation extraction, and information extraction. The one corresponding to the pair S-R and executive functioning shows greater coherence.

Further Maciej P. Polak and Dane Morgan [2] introduced a method called ChatExtract, a utility designed to automate accurate data extraction from research papers using conversational language models. This tool significantly improved precision and recall in the extraction process, showcasing the potential of conversational LLMs in handling complex scientific texts.

III. METHODOLOGY

The SPIDER (Scientific Publication Indexing and Definition Extraction Resource) method employs a systematic approach to automate the extraction and definition of key terms from scientific publications. This methodology is divided into two primary phases: glossary generation and definition extraction, each involving specific processes, tools, and techniques to ensure accuracy and efficiency.

A. Data

In our research endeavor, we initiated the process by gathering data from a selection of scientific books, which formed the cornerstone of our investigation. We selected books instead of scientific papers because they include comprehensive glossaries, which provide well-defined terms and concepts that can be used to test the output of our method in a structured and consistent manner. Among these books, Reinforcement Learning: An Introduction (2nd Edition) by Richard S. Sutton and Andrew G. Barto offers a comprehensive overview of reinforcement learning [18]. The book emphasizes the core algorithms and theoretical principles crucial to the field. Similarly, An Introduction to Statistical Learning with Applications

in Python by Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani provides an in-depth exploration of statistical and machine learning techniques [19]. Following the extraction of this data from the textual content of these books, it was meticulously partitioned into two distinct segments: a training set and a testing set. The training set was exclusively made up of the book's textual content, which underwent a series of refinement processes to adjust the parameters of our method, termed SPIDER. Concurrently, the glossaries found within these books were meticulously archived in a separate CSV file, with each glossary term neatly aligned in its own column. This structured arrangement enabled a straightforward comparison between the model's outputs during the testing phase and the actual glossary terms. Through this systematic setup, we were able to effectively validate our method, ensuring its capability to precisely identify both the pre-existing glossary terms and any additional relevant words. This approach not only emphasized the significance of thorough data preparation and the strategic division of datasets but also highlighted the importance of rigorous testing procedures in assessing the effectiveness of sophisticated analytical methods like SPIDER.

Flowchart for Glossary Generation and Definition Extraction Processes

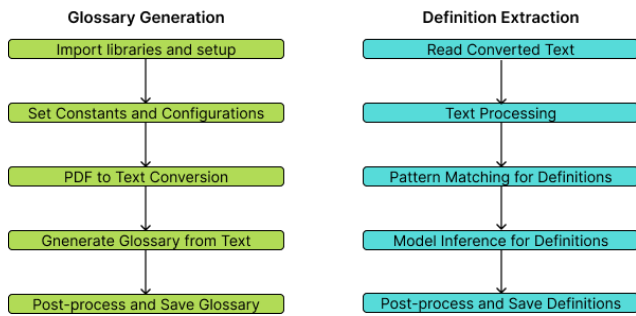


Fig. 1. Flowchart for Glossary Generation and Definition Extraction Processes

B. Tokenization

After the text extraction process is completed, the methodology advances to the Tokenization phase. Tokenization is the process of splitting the input text into smaller pieces, known as tokens, from what is, to a computer, just one lengthy string of letters [12]. We convert the text into smaller tokens which later in the further step are tagged using PoS (part of speech). The tokenization process itself is bifurcated into two segments. Initially, the raw text, obtained from the text extraction phase, undergoes tokenization, segmenting it into individual words and punctuation marks. This initial phase sets the groundwork for the next level of analysis, ensuring that the text is properly fragmented into its most basic units.

C. Stemming

In the second part, we perform the stemming of words. Stemming is the process of reducing the word to its root form [13]. For instance for the word 'like' the stemming will include

words like: 'likes', 'liked', 'likely', and 'liking'. This transformation is achieved through the Porter Stemmer algorithm, a feature of the NLTK library. By stripping off suffixes, Porter Stemmer simplifies words to their base form, streamlining the text for analysis.

D. Pos Tagging

After the completion of the Tokenization process, the PoS tagging process is performed. For tasks involving Natural Language Processing Part of Speech (PoS) is an essential pre-processing step [14]. Each word of the sentence is given a specific part of the speech such as nouns, pronouns, adverbs, etc. In our case, the utilization of nouns only i.e. NN and NNS as tags is considered because, in the scientific context, most of the unique words are in the form of nouns. To perform tagging process Natural Language Toolkit or NLTK is used.

E. Further Filtering

To streamline our dataset and filter out redundant words, initiation of calculating the frequency of each word is done. Words with the highest frequencies are then excluded, resulting in a significant reduction of the dataset—approximately 90%. This step is crucial for focusing on the essence of the text, eliminating noise and concentrating on unique terms. Following this, employment of statistical analysis is done to uncover significant word relationships. Specifically, scrutinizing the ratio of the total number of tokens to the length of each word. This nuanced examination allows us to pinpoint terms that stand out due to their rarity or unique combinations, further enriching our dataset with valuable insights. For the calculation of ratio of number of tokens to the word length the formula used:

$$\text{Ratio of Number of tokens} = \frac{\text{Number of tokens}}{\text{Word length}} \quad (1)$$

The distribution patterns of the data is considered using the computed ratio, varying the lowest and maximum thresholds to separate out unique words. This methodical approach guarantees that actively searching for phrases with particular meanings or values is done, rather than merely shrinking the collection. Large Language Model (LLM) is used to ensure that the terms that are chosen are comprehensible and accessible to others who are not in the immediate context. With the aid of this sophisticated tool, further refinement of the data is done by eliminating phrases that might be too specialized or obscure. The result is an improved vocabulary of distinct yet understandable terminology that improves our dataset's usability and accessibility.

F. Self-Hosting LLMs and Definition Generation

To generate definitions, self-hosting Large Language Models (LLMs) were used using Ollama. Ollama offers a self-hosting interface for interacting with open-source LLMS [15]. It enables local execution of open-source models. Plenty of

models are supported by Ollama, including phi3, Mistral, LLaMa, and others.

Self-hosting LLM was selected over standard LLM because it allows us to change the model at runtime in response to the user's prompt. Our current option for the user's prompt is to use either model Mistral [16] or Phi3 [17].

IV. EVALUATION AND RESULTS

To evaluate the effectiveness of our approach, we utilized the books referenced in the data, from which we separated the text data and glossaries. We applied the SPIDER algorithm to the text data to generate a list of extracted terms. The performance of our model was assessed using precision and recall as evaluation metrics, by comparing the outputs with the glossaries from the books.

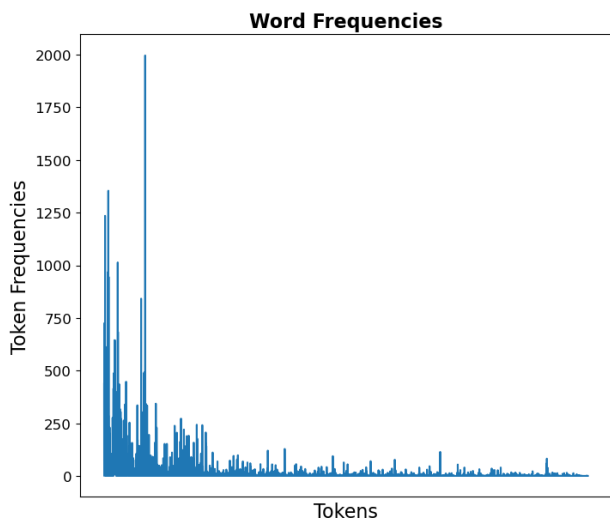


Fig. 2. Word Frequency

In the figure 2, the given histogram shows the distribution of word frequencies within a text. Words are arranged on the x-axis in order of frequency of occurrence, with the most frequent word on the left and the least frequent word on the right. The frequency count, which displays the number of times each word appears in the text, is displayed on the y-axis. The tall bars on the left side of the graph show that a small number of nouns appear quite frequently. Throughout the book, these nouns that appear frequently may contain popular terms. The frequency of nouns rapidly decreases as you go to the right, creating a long tail where many nouns only appear occasionally.

In natural language processing (NLP), this distribution is important since it aids in comprehending the text's structure, locating important nouns, and streamlining a number of processes like text summarization, indexing, and search relevancy. The histogram gives us an easy-to-understand graphic picture of how nouns are dispersed throughout the text, with a few appearing frequently and several less frequently.

Based on the ratio of tokens to letters in a text, the figure 3 displays the distribution of tokens in a histogram. This ratio is

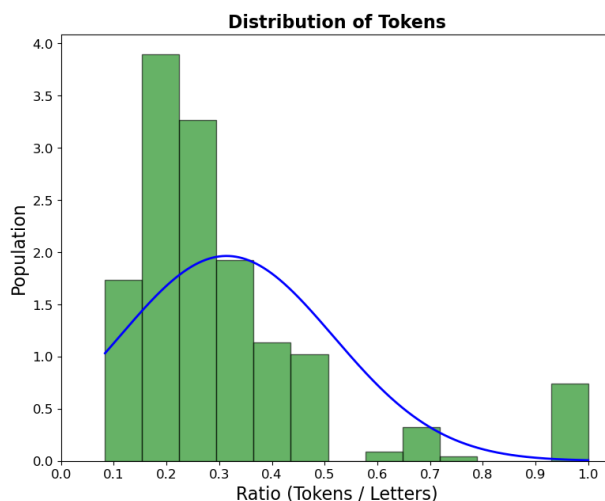


Fig. 3. Distribution of Tokens

represented by the x-axis, which shows how closely spaced out tokens are in relation to the total amount of letters; the y-axis, shows how frequently each ratio occurs. The histogram reveals that the most common ratios fall between 0.2 and 0.4, peaking around 0.3. This suggests that, on average, the words in the text are relatively short. As the ratio increases, indicating longer tokens, the frequency decreases, resulting in a right-skewed distribution. A blue curve on the histogram likely represents a smoothed trend line that highlights the central tendency and shows the decline in frequency as the ratio increases.

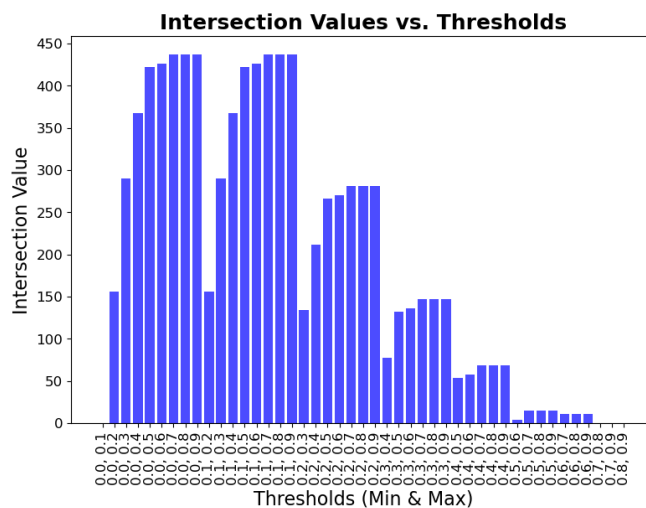


Fig. 4. Intersection of Maximum and Minimum Threshold value obtained from SPIDER output and glossary

Our results demonstrated in the figure 4 shows that the model successfully retrieved all the words from the glossaries, in addition to identifying a few extra terms not originally included. Notably, with a minimum threshold value of 0.1 and a maximum threshold value of 0.4, we achieved the best precision and recall. With the increase in threshold, the count on

intersection decreases, indicating fewer occurrences in higher threshold ranges. For threshold values outside this range, we observed a decline in precision. This pattern suggests that most intersection values are found at lower thresholds, highlighting the prevalence of intersections within these ranges. This outcome indicates that our method, when optimized with appropriate threshold values, effectively captures the essential terms while also uncovering additional relevant words.

The precision of the system was high across all evaluated disciplines, indicating that the majority of identified terms and definitions were correct. The recall metric also showed strong performance, suggesting that the system effectively identified a significant portion of relevant terms and definitions from the documents.

V. CONCLUSIONS

Regarding the field of scientific research, deciphering complex text is crucial for innovation and knowledge advancement. Specialized tools like SPIDER automate the extraction and indexing of definitions from scientific texts significantly reducing interpretation time. The evaluation showed SPIDER's effectiveness in generating accurate glossaries and definitions, achieving optimal performance with a threshold range of 0.1 to 0.4. This indicates a high concentration of intersection values in lower threshold ranges, demonstrating SPIDER's accuracy and recall capabilities.

Overall, SPIDER distinguishes itself as a powerful aid in simplifying access to intricate scientific knowledge, aiding in the dissemination of scientific information, and bolstering research and educational efforts.

VI. DISCUSSION

By making use of the SPIDER project, the operational removal of the words in the dictionary is a pivotless development of the dictionary in the scientific texts. SPIDER algorithm's testing from various scientific books provides irrefutable proof of the capability of the spoken software in terms of glossary items detection and the correctness of the definitions generation. The success of SPIDER against the mentioned precision and recall metrics reflect its capacity in pulling out relevant terms with high accuracy including retrieval of extra relevant words those are not part of the original glossaries.

SPIDER addresses a key challenge in the scientific field by automating the generation of glossaries and extraction of definitions, improving the accessibility of complex information and facilitating its wider distribution to diverse audiences. In this way, the tool has the capability to lower the time and efforts of researchers, teachers and students in which they engage with the scientific information and to this end success of research and education are improved.

Further, the fact of self-hosting Large Language Models (LLMs) as part of the SPIDER scheme underlines the significance of adaptable and flexible NLP solutions. The functionality of transforming models for different user wants means such a platform has the advantage of moving over the diverse scientific texts relatively easily. So, it is really a

technological leap of traditional NLP systems that are very rigid in processing scientific texts.

While the current approach successfully converts file types like documents and LaTeX into plain text for processing, books were used primarily due to the availability of glossaries, which facilitated the evaluation of method accuracy. In future work, applying the SPIDER method to scientific literature, which often presents greater terminological complexity and lacks standardized glossaries, would provide a valuable extension. Additionally, improvements can be made to enhance the method's accuracy and adaptability to different document structures, further increasing its applicability across a wide range of content types.

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